



An Introduction to Rigid Analytic Geometry

Yusuf Emre Baran

Advisor: Enis Kaya

Course Coordinator: Deniz Yılmaz

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1 Introduction

Classical analytic geometry relies heavily on the topological properties of the real and complex fields. However, when working over a field k equipped with a complete, non-trivial, non-Archimedean valuation, standard topological methods fail because the metric space induced by the valuation is totally disconnected. To perform analytic geometry over such fields, one must adopt the framework of Rigid Analytic Geometry, pioneered by John Tate. This report traces the foundational construction of rigid analytic varieties. We begin by reviewing affine varieties and schemes. Then we study the Tate algebra of convergent power series and its quotients, the k -affinoid algebras. By establishing the maximal ideal spectrum as our geometric space, we define affinoid subdomains and construct a specialized Grothendieck topology. Through a systematic refinement from the weak to the strong G -topology, we demonstrate how these local affinoid building blocks can be glued together to form global rigid analytic varieties, ultimately providing a robust framework for non-Archimedean analysis.

The main reference is [BGR84], which provides a complete treatment of non-archimedean analysis and contains everything in Chapter 3. We assume familiarity with the basics of non-archimedean valued fields (as a special case, one can look at [Gou20]), field extensions, and sheaf theory (for G -topologies, we also assume some knowledge of basic category theory, [Lei14] is a nice resource for this). Of course, we also assume familiarity with varieties ([CLO25] is a nice introductory textbook for a beginner). We do not discuss the technical details of schemes; our focus on them is a bridge to rigid analytic varieties. For detailed treatments of schemes, one can look at [Har77] or [Qin06], and [EH00] for a more geometric treatment.

2 Schemes

2.1 Review of Affine Varieties

In this section, we recall the foundational results of classical algebraic geometry that will lead us to the theory of schemes, which in turn will guide us on our way to defining rigid analytic spaces.

One is accustomed to classical algebraic geometry as a correspondence between geometry and algebra in affine space. To be more precise, let k be an algebraically closed field and $\mathbb{A}^n$ be the affine n -space over k . One then establishes a link between subsets of $\mathbb{A}^n$ and ideals in the polynomial ring $R = k[x_1, \dots, x_n]$ via two inclusion-reversing maps Z and I defined by

$$Z(S) = \{p \in \mathbb{A}^n \mid f(p) = 0 \text{ for all } f \in S\}$$

for any subset $S \subset R$ and

$$I(X) = \{f \in R \mid f(p) = 0 \text{ for all } p \in X\}$$

for any $X \subset \mathbb{A}^n$. By building on Hilbert's Nullstellensatz, one obtains a bijective correspondence between affine varieties in $\mathbb{A}^n$ and prime ideals in $k[x_1, \dots, x_n]$:

$$\{\text{affine varieties in } \mathbb{A}^n\} \xrightleftharpoons[Z]{I} \{\text{prime ideals in } k[x_1, \dots, x_n]\}. \quad (2.1)$$

2.2 Transitioning to Schemes

If one is familiar with commutative algebra, then one can replace the right-hand side of (2.1) by finitely generated, reduced k -algebras, obtaining

$$\{\text{affine varieties in } \mathbb{A}^n\} \longleftrightarrow \{\text{affine } k\text{-domains}\}. \quad (2.2)$$

The upshot of this replacement is that we can now see clearly that the right-hand side is just a ring with some properties. So one naturally wonders whether it is possible to obtain a similar correspondence for any ring, not just for finitely generated reduced k -algebras.

Of course, the question is then what happens to the left-hand side. In the next section, we will answer this question.

To get an idea of how we might proceed with our goal, let $X \subset \mathbb{A}^n$ be an affine variety and let $k[X] = k[x_1, \dots, x_n]/I(X)$ be its coordinate ring. Then by Nullstellensatz, every point $p = (a_1, \dots, a_n) \in X$ corresponds uniquely to the maximal ideal $\mathfrak{m}_p = (x_1 - a_1, \dots, x_n - a_n)$ in $k[x_1, \dots, x_n]$. Consider the evaluation homomorphism

$$\phi_p : k[x_1, \dots, x_n] \rightarrow k, \quad f \mapsto f(p). \quad (2.3)$$

One can show that $\ker(\phi_p) = \mathfrak{m}_p$. Then $f \in I(X)$ implies that $f \in \mathfrak{m}_p$ simply by definition of $I(X)$. Thus $\mathfrak{m}_p \supset I(X)$. Moreover, the correspondence theorem for rings tells us that there is a one-to-one correspondence between the maximal ideals of $k[X]$ and the maximal ideals of $k[x_1, \dots, x_n]$ containing $I(X)$. Combining this with the correspondence mentioned at the beginning, we obtain

$$\{\text{points of } X\} \longleftrightarrow \{\text{maximal ideals of } k[x_1, \dots, x_n] \text{ containing } I(X)\}. \quad (2.4)$$

Next, let $\mathfrak{p} \supset I(X)$ be a prime ideal that is not maximal. By (2.1), this prime ideal corresponds to some affine variety, say Y . Because $I(X)$ is a prime ideal and Z is inclusion-reversing, we have $Y = Z(\mathfrak{p}) \subset Z(I(X)) = X$. So we obtain another correspondence

$$\{\text{subvarieties of } X\} \longleftrightarrow \{\text{prime ideals of } k[x_1, \dots, x_n] \text{ containing } I(X)\}. \quad (2.5)$$

In fact, (2.4) is a special case of (2.5) because a point $p \in X$ can be seen as the subvariety determined by the set of equations $x_i - a_i = 0$ for $1 \leq i \leq n$, and maximal ideals are prime. In particular, the prime ideals in (2.5) cannot be maximal.

Keeping in mind that the geometry of X is encoded in its coordinate ring $k[X]$, we employ the correspondence theorem one last time to get

$$X \cong \text{prime ideals of } k[X], \quad (2.6)$$

where our goal in this section is to give a precise meaning to $\cong$. Before we start that, it is worth taking a minute to summarize what happened. We realize that we no longer need the scaffold of the ambient affine space $\mathbb{A}^n$ to define our geometry. The ring $k[X]$ is enough

to encode geometric information via prime ideals. By stripping away the requirement that our ring must be a finitely generated k -algebra over an algebraically closed field, we can apply this same geometric intuition to any commutative ring R . In other words, we can do geometry by starting from an algebraic point of view. We now discuss this framework.

2.3 The Spectrum of a Ring

Motivated by the right-hand side of (2.6), we make the following definition:

Definition 2.1. Let R be a ring. The spectrum of R , denoted $\text{Spec } R$, is the set of all prime ideals of R

$$\text{Spec } R = \{\mathfrak{p} \subset R \mid \mathfrak{p} \text{ is a prime ideal}\}.$$

Of course, this definition by itself doesn't help us reach something geometric. Our first hurdle is thus to find our way into geometry. For that, we again turn to our experience of varieties. Given an ideal $I \subset R$, let

$$V(I) := \{\mathfrak{p} \in \text{Spec } R \mid \mathfrak{p} \supset I\}.$$

At first sight, this definition of $V(I)$ doesn't seem related to either of the operators I or Z we know from varieties at all. We claim that $V(I)$ is the analog of $Z(I)$. The key observation here is the mantra of modern algebraic geometry: elements of a ring can be viewed as functions. To make it more concrete, recall that the kernel of the evaluation homomorphism (2.3) is $\mathfrak{m}_p$, which is to say that $f \in k[x_1, \dots, x_n]$ vanishes at p if and only if $f \in \mathfrak{m}_p$. In the case of an arbitrary ring R , we can define the value of an element $f \in R$ at a point $\mathfrak{p}^1 \in \text{Spec } R$ as its image in the quotient $R/\mathfrak{p}$:

$$f(p) := f \pmod{\mathfrak{p}} \in R/\mathfrak{p}.$$

While $f \pmod{\mathfrak{p}}$ naturally lives in the integral domain $R/\mathfrak{p}$, we embed this domain into its field of fractions to satisfy our geometric intuition that functions should take values in a field. We define the residue field $\kappa(\mathfrak{p})$ of $\text{Spec } R$ at $\mathfrak{p}$ to be the quotient field of $R/\mathfrak{p}$. So an element of $f \in R$ can be viewed as a function where $f(\mathfrak{p}) \in \kappa(\mathfrak{p})$ to be the image

¹It might be helpful to use another notation for $\mathfrak{p}$ when it is viewed as an element of $\text{Spec } R$. However, we trust that the context will make it clear whether $\mathfrak{p}$ is viewed as a prime ideal of R or as an element of $\text{Spec } R$.

of f under the composition of the canonical maps

$$R \longrightarrow R/\mathfrak{p} \hookrightarrow \kappa(\mathfrak{p}).$$

Hence an element $f \in R$ is now a function on $\text{Spec } R$ where the target fields change with $\mathfrak{p}$, namely we have

$$f : \text{Spec } R \longrightarrow \prod_{\mathfrak{p} \in \text{Spec } R} \kappa(\mathfrak{p}), \quad (2.7)$$

which reminds one the notion of a bundle. We can now see clearly how $V(I)$ relates to $Z(I)$. One has

$$\{\mathfrak{p} \in \text{Spec } R \mid \mathfrak{p} \supset I\} = \{\mathfrak{p} \in \text{Spec } R \mid f(\mathfrak{p}) = 0 \text{ for all } f \in I\}. \quad (2.8)$$

This follows because for any $f \in R$, one has

$$f(\mathfrak{p}) = 0 \text{ in } \kappa(\mathfrak{p}) \iff f \equiv 0 \pmod{\mathfrak{p}} \iff f \in \mathfrak{p}.$$

In other words, $V(I)$ is the intersection of points in $\text{Spec } R$ where all the elements of I vanish. Also note that if $S \subset R$ is a subset such that $(S) = I$, then $V(S) = V(I)$. So we can consider V as a map on the ideals of R without loss of generality.

Now that we have begun to see that $\text{Spec } R$ is a promising candidate for our theory, let's make it more polished by giving it a topology that is suitable for our purposes.

2.4 The Zariski Topology on $\text{Spec } R$

The following observation allows us to define a topology on $\text{Spec } R$:

Proposition 2.2. Let R be a ring. Then V satisfies the following properties:

- (i) For any pair of ideals I, J of R , we have $V(I) \cup V(J) = V(I \cap J)$.
- (ii) Let $(I_\lambda)_\lambda$ be a family of ideals of R . Then $\bigcap_\lambda V(I_\lambda) = V(\sum_\lambda I_\lambda)$.
- (iii) $V(R) = \emptyset$ and $V(0) = \text{Spec}(R)$.

Proof. We show (i) and (ii), the last one is obvious. If $\mathfrak{p}$ contains I or J , it clearly contains $I \cap J$. Conversely, if $\mathfrak{p}$ contains $I \cap J \supset IJ$, then $IJ \subset \mathfrak{p}$. Hence $I \subset \mathfrak{p}$ or $J \subset \mathfrak{p}$ as $\mathfrak{p}$ is prime, proving (i). Finally, (ii) is clear since $\mathfrak{p}$ contains each I_λ if and only if it contains $\sum_\lambda I_\lambda$. □

By this proposition, we see that the sets of the form $V(I)$ form the closed sets of a topology on $\text{Spec } R$, called the Zariski topology. Additionally, this proposition gives an alternative perspective on the definition of $V(I)$. Even though the right-hand side of (2.7) has no topology (the fields $\kappa(\mathfrak{p})$ have no topology), each $\kappa(\mathfrak{p})$ has one thing in common, called the zero element. So if we treat the function f as a continuous map and assume for a moment that the codomain has a Hausdorff topology, then the inverse image $f^{-1}(0)$ should be a closed set in $\text{Spec } R$. We are then led to defining $V(I)$ as in (2.8).

After giving $\text{Spec } R$ a topology, we turn our attention to understanding its local behavior. We consider the case where $V(I)$ is determined by a single element $f \in R$, i.e., $I = (f)$. Recall that in classical geometry, $Z(f)$ defines a hypersurface, and its complement is the region where f is invertible. This motivates the following definition:

Definition 2.3. For any $f \in R$, the basic open set $D(f)$ is defined as

$$D(f) := \text{Spec } R \setminus V(f) = \{\mathfrak{p} \in \text{Spec } R \mid f \notin \mathfrak{p}\}.$$

Loosely, each $D(f)$ encodes the behavior of f near points where it is nonzero. So one might expect that these open sets form a basis for the Zariski topology on $\text{Spec } R$. Indeed, this is the case, as the next proposition shows:

Proposition 2.4. The basic open sets $D(f)$ form a basis for the Zariski topology on $\text{Spec } R$.

Proof. Let U be an open set in $\text{Spec } R$ such that $U = \text{Spec } R \setminus V(I)$ for some ideal $I \subset R$. Remembering that $I = \sum_{f \in I} (f)$, Proposition (2.2) gives

$$U = \text{Spec } R \setminus V(I) = \text{Spec } R \setminus \bigcap_{f \in I} V(f) = \bigcup_{f \in I} (\text{Spec } R \setminus V(f)) = \bigcup_{f \in I} D(f).$$

Moreover, each basic open set is clearly open. This finishes the proof. $\square$

To see how localization interacts with our new topology, let R_f be the localization of R at the element f . In the usual notation, this means that $R_f = S^{-1}R$, where $S = \{f^n \mid n \geq 0\}$. It is a standard result that the prime ideals of R_f are in one-to-one correspondence with the prime ideals of R that do not intersect S . Since a prime ideal $\mathfrak{p}$

intersects S if and only if $f \in \mathfrak{p}$, we obtain a correspondence

$$\mathrm{Spec} R_f \longleftrightarrow \{\mathfrak{p} \in \mathrm{Spec} R \mid f \notin \mathfrak{p}\} = D(f). \quad (2.9)$$

The next result shows that the above correspondence also preserves topology.

Proposition 2.5. Let R be a ring and let $f \in R$. Let $\phi : R \rightarrow R_f$ be the localization map. The induced map

$$\phi^* : \mathrm{Spec} R_f \longrightarrow \mathrm{Spec} R, \quad \mathfrak{q} \longmapsto \phi^{-1}(\mathfrak{q})$$

is a homeomorphism of $\mathrm{Spec} R_f$ onto the basic open set $D(f) \subseteq \mathrm{Spec} R$, where $\mathrm{Spec} R_f$ is given the Zariski topology.

Proof. As noted above, a prime ideal $\mathfrak{p} \subset R$ intersects S if and only if $f \in \mathfrak{p}$. Taking the negation, $\mathfrak{p}$ does not intersect S if and only if $f \notin \mathfrak{p}$. The set of such prime ideals is exactly $D(f) = \{\mathfrak{p} \in \mathrm{Spec} R \mid f \notin \mathfrak{p}\}$. This shows that the map ϕ^* is injective and its image is $D(f)$.

Next, we show that ϕ^* is continuous. Let $V(I)$ be a closed set in $\mathrm{Spec} R$. Then

$$(\phi^*)^{-1}(V(I)) = \{\mathfrak{q} \in \mathrm{Spec} R_f \mid \phi^{-1}(\mathfrak{q}) \supseteq I\}$$

By the property of localization, $\phi^{-1}(\mathfrak{q}) \supseteq I$ if and only if $\mathfrak{q} \supseteq I \cdot R_f$. Thus:

$$(\phi^*)^{-1}(V(I)) = V(I \cdot R_f)$$

Since $I \cdot R_f$ is an ideal in R_f , this set is closed in $\mathrm{Spec} R_f$. Hence ϕ^* is continuous.

Finally, we show that ϕ^* is an open map onto $D(f)$. It suffices to check openness on basic open sets as they form a basis for the Zariski topology. A basic open set in $\mathrm{Spec} R_f$ is of the form $D(g/f^n)$ for some $g \in R$ and $n \geq 0$. On the other hand, we have $D(g/f^n) = D(g/1)$ in R_f . An element $\mathfrak{q} \in \mathrm{Spec} R_f$ is in $D(g/1)$ if and only if $g/1 \notin \mathfrak{q}$. This is equivalent to saying the corresponding prime $\mathfrak{p} = \phi^{-1}(\mathfrak{q})$ satisfies $g \notin \mathfrak{p}$ and $f \notin \mathfrak{p}$. The condition $g \notin \mathfrak{p}$ and $f \notin \mathfrak{p}$ is equivalent to $fg \notin \mathfrak{p}$. Therefore, the image $\phi^*(D(g/1))$ is $D(fg) \subseteq \mathrm{Spec} R$. Since $D(fg)$ is a basic open set in $\mathrm{Spec} R$ and $D(fg) \subseteq D(f)$, it is open in the subspace topology of $D(f)$. This completes the proof. $\square$

We will frequently identify $\text{Spec } R_f$ with $D(f)$ thanks to this result, which tells us that every basic open set $D(f)$ of an affine scheme is itself an affine scheme, namely $\text{Spec } R_f$.

Remark 2.6. This result mirrors a fundamental property of manifolds: any open subset of a manifold is itself a manifold. In our new framework, while it is true that any open subset of a scheme is a scheme, the sets $D(f)$ are special because they are guaranteed to be affine schemes. This makes $D(f)$ the algebraic equivalent of a coordinate chart. Just as a manifold is a space built by gluing together pieces that look like $\mathbb{R}^n$, a general scheme is a space built by gluing together open sets that look like $\text{Spec } R$.

2.5 Structure Sheaves

Before we restrict our attention to schemes, it is insightful to discuss sheaves in general and their role in geometry.

To understand why $\mathcal{O}_X$ is called the structure sheaf (the emphasis is on the word structure), it is again instructive to look at the category of manifolds. A topological space M can be many things simultaneously; the sheaf is what forces it to be a specific type of geometric object. To understand what we mean by that, let M be a d -dimensional topological manifold. We define the sheaf of smooth functions $\mathcal{C}_M^\infty$ by assigning to each open set $U \subset M$ the ring

$$\mathcal{C}_M^\infty(U) := \{f : U \rightarrow \mathbb{R} \mid f \text{ is smooth}\}.$$

One can check that $\mathcal{C}_M^\infty$ is indeed a sheaf on M . It is also possible to define another sheaf $\mathcal{C}_M^\omega$ by requiring the map f above to be real analytic. Even if the underlying set of points M and the topology are identical, $(M, \mathcal{C}_M^\infty)$ and $(M, \mathcal{C}_M^\omega)$ are different geometric objects. In fact, we can define a smooth manifold formally as a pair $(M, \mathcal{C}_M^\infty)$ that is locally isomorphic to $(\mathbb{R}^d, \mathcal{C}_{\mathbb{R}^d}^\infty)$. This was indeed the early definition of a manifold². It can be shown that by accepting that definition, the transition functions $\psi_\beta \circ \psi_\alpha^{-1}$ are forced to be smooth, giving back the definition we are familiar with.

What we have been trying in this subsection is that there is at least some justification in calling the sheaf that gives an identity to a scheme the structure sheaf. If the

²In fact, some books still define manifolds using sheaves, see for example [Wed16].

reader found the discussion obscure and vague, they can safely skip it.

Now that we have some understanding of the topology of $X^3 = \operatorname{Spec} R$, one next studies functions on X in alignment with core principles of understanding a new object. This then requires specifying a sheaf of functions as we have seen in the previous subsection. Again, we turn to varieties for help.

Recall that geometric properties of an affine variety Y is mirrored by an algebraic property of its coordinate ring $k[Y]$, and most importantly, $k[Y]$ is precisely the ring of functions defined everywhere on Y . We saw in (2.7) that elements of R are functions on X . We want nothing more, i.e., we want the ring of functions defined on X to be exactly R . In the notation of sheaves, we wish to have

$$\mathcal{O}_X(X) = R.$$

After declaring the functions on X , we then ask what should be the functions defined on an open set $U \subset X$. We first consider basic open sets $D(f)$, where f is guaranteed to be non-vanishing. In the usual algebra-geometry dictionary, if a function f never takes the value zero on a neighborhood, its reciprocal $1/f$ should be a perfectly valid function on that neighborhood. Algebraically, the most efficient way to allow $1/f$ is to localize R at f . Hence, we are led to define

$$\mathcal{O}_X(D(f)) := R_f$$

for basic open sets. Note that if $D(g) \subset D(f)$, then some power of g is a multiple of f . So we can let the restriction map $\rho_{D(f)D(g)}$ to be the localization map $R_f \rightarrow R_{fg} = R_g$. Then it is a classic result in sheaf theory that $\mathcal{O}_X$ extends uniquely to a sheaf on X (see [EH00]).

We are now ready to define schemes. One first talks about affine schemes, which are a pair $(X, \mathcal{O}_X)$ just constructed. Then the definition of a scheme follows:

Definition 2.7. Let X be a topological space and let $\mathcal{O}_X$ be a sheaf of rings on X such that $\mathcal{O}_{X,x}$ is a local ring for each $x \in X$. The pair $(X, \mathcal{O}_X)$ is called a scheme if there exists an open covering $\{U_i\}$ of X such that $(U_i, \mathcal{O}_X|_{U_i})$ is isomorphic to an affine scheme.

By $(U_i, \mathcal{O}_X|_{U_i})$ being isomorphic to an affine scheme, we mean that there is a ring

³From now on, we will write X in place of $\operatorname{Spec} R$. This comes from our inclination of thinking $\operatorname{Spec} R$ as a geometric object and $\operatorname{Spec} R$ denoting the prime ideals of R does not serve us in that goal.

R_i and a pair of maps $(f, f^\sharp)$ such that $f : U_i \rightarrow \operatorname{Spec} R_i$ is a homeomorphism and $f^\sharp : \mathcal{O}_{\operatorname{Spec} R_i} \rightarrow f_* \mathcal{O}_X|_{U_i}$ is an isomorphism of sheaves, where $f_* \mathcal{O}_X|_{U_i}$ is the pushforward of $\mathcal{O}_X|_{U_i}$ by f defined by

$$f_* \mathcal{O}_X|_{U_i}(V) = \mathcal{O}_X|_{U_i}(f^{-1}(V))$$

for any open $V \subset \operatorname{Spec} R_i$.

3 Rigid Analytic Varieties

Using the ideas and methods we have learned from schemes, we now begin the main discussion of this report. The question that motivates us is how one goes about defining analytic varieties when the ground field k is equipped with a non-archimedean valuation. While scheme theory is successful in generalizing the notion of a variety and accomplishes much more, it is blind to nonarchimedean fields because of their totally disconnected nature. We now begin building a framework that allows for non-archimedean-valued fields.

3.1 Tate Algebras

Just as in the study of varieties and schemes, we need a building block for our new theory. This building block is again algebraic in nature, but much closer to allowing us to consider fields with a non-archimedean valuation. For the rest of this report, a field k at least means a complete field with respect to a non-trivial non-archimedean valuation, which is not assumed to be algebraically closed. Here is the definition of our building block.

Definition 3.1. Let k be a field. The Tate algebra $T_n(k) = k\langle \xi^4_1, \dots, \xi_n \rangle$ is the subring of the ring of formal power series $k[[\xi_1, \dots, \xi_n]]$ consisting of convergent elements. If k is understood, we write T_n instead of $T_n(k)$.

Since a series $\sum_{\nu=0}^{\infty} a_\nu$ converges if and only if $|a_\nu| \rightarrow 0$, we have

$$T_n = \left\{ \sum_{\nu} a_\nu \xi^\nu : |a_\nu| \rightarrow 0 \right\},$$

⁴We may also use z_i as an indeterminate in the following.

where we employ the usual multi-index notation, i.e., $\nu = (\nu_1, \dots, \nu_n)$ and $\xi^\nu = \xi_1^{\nu_1} \dots \xi_n^{\nu_n}$. So $a_\nu = a_{\nu_1 \dots \nu_n}$ and $|a_\nu| \rightarrow 0$ if $|\nu| = \nu_1 + \dots + \nu_n \rightarrow \infty$.

Observe that since $|a_\nu| \rightarrow 0$, we have $|a_\nu| \geq \varepsilon$ for finitely many multi-indices ν for any $\varepsilon > 0$. This means that $\max_\nu$ is attained at some a_ν for any $f = \sum_\nu a_\nu \xi^\nu$ in T_n . So we can define

$$|f| = \max_\nu |a_\nu|.$$

Theorem 3.2. The function $|\cdot| : T_n \rightarrow \mathbb{R}_{\geq 0}$ defined above is a norm on T_n , called the Gauss norm, and makes it a k -Banach algebra.

Proof. Let $f = \sum a_\nu \xi^\nu$ and $g = \sum b_\nu \xi^\nu$ be elements of T_n , and $\lambda \in k$. We have $|f| = 0$ if and only if $\max_\nu |a_\nu| = 0$, which implies $a_\nu = 0$ for all ν , hence $f = 0$. Next, one has $|\lambda f| = \max_\nu |\lambda a_\nu| = |\lambda| \max_\nu |a_\nu| = |\lambda| |f|$. To show the ultrametric inequality, we use the valuation on k . Using $\max_\nu (\max(|a_\nu|, |b_\nu|)) = \max(\max_\nu |a_\nu|, \max_\nu |b_\nu|)$, one gets

$$|f + g| = \max_\nu |a_\nu + b_\nu| \leq \max_\nu (\max(|a_\nu|, |b_\nu|)) = \max(|f|, |g|).$$

Before we show that the norm is multiplicative, we first show that it is submultiplicative. Let $fg = \sum c_\nu z^\nu$, where $c_\nu = \sum_{\mu+\lambda=\nu} a_\mu b_\lambda$. By the ultrametric inequality in k , one gets

$$|c_\nu| = \left| \sum_{\mu+\lambda=\nu} a_\mu b_\lambda \right| \leq \max_{\mu+\lambda=\nu} |a_\mu b_\lambda| = \max_{\mu+\lambda=\nu} (|a_\mu| |b_\lambda|).$$

Since $|a_\mu| \leq |f|$ and $|b_\lambda| \leq |g|$ for all μ, λ , we have $|c_\nu| \leq |f| |g|$. Taking the maximum over ν yields $|fg| \leq |f| |g|$.

To see that the norm is multiplicative, we may suppose that f and g are nonzero. So there exist $\alpha, \beta \in k$ such that $|\alpha| = |f|^{-1}$ and $|\beta| = |g|^{-1}$ by the paragraph before the theorem. Define $f' = \alpha f$ and $g' = \beta g$. Note that $|f'| = 1$ and $|g'| = 1$. If we can prove that $|f'g'| = 1$, then homogeneity yields

$$|\alpha\beta fg| = 1 \implies |\alpha| |\beta| |fg| = 1 \implies \frac{1}{|f|} \frac{1}{|g|} |fg| = 1 \implies |fg| = |f| |g|$$

Thus, it suffices to prove the theorem for $|f| = 1 = |g|$.

Let $k^\circ = \{x \in k : |x| \leq 1\}$ be the valuation ring of k , and let $k^\vee = \{x \in k : |x| < 1\}$ be its maximal ideal. Denote by $\tilde{k} = k^\circ / k^\vee$ the residue field of k . For any $h = \sum c_\nu \xi^\nu \in T_n$

with $|h| \leq 1$, we have $c_\nu \in k^\circ$ for all ν . Hence we can define the reduction $\tilde{h} \in \tilde{k}[\xi_1, \dots, \xi_n]$ by reducing each coefficient $c_\nu \bmod k^\vee$. Because $|c_\nu| \rightarrow 0$, only finitely many coefficients have absolute value 1, which implies that $\tilde{h}$ is indeed in $\tilde{k}[z_1, \dots, z_n]$. Because $|f| = 1$, we have $|a_\nu| = 1$ for some ν . Hence $\tilde{f} \neq 0$. Similarly, $\tilde{g} \neq 0$. Then $\widetilde{fg} = \tilde{f}\tilde{g}$ implies that $\widetilde{fg} \neq 0$ because $k^\sim[z_1, \dots, z_n]$ is an integral domain. But if $\widetilde{fg} \neq 0$, the product fg has at least one coefficient that has absolute value 1. So $|fg| \geq 1$. We already know that $|fg| \leq |f||g| = 1$. This shows that $|fg| = 1 = |f||g|$.

Finally, let $\{f_m\}_{m=1}^\infty$ be a Cauchy sequence in T_n with $f_m = \sum_\nu a_{m,\nu} \xi^\nu$. For any fixed ν , we have

$$|a_{m,\nu} - a_{l,\nu}| \leq |f_m - f_l|.$$

This shows that $\{a_{m,\nu}\}_{m=1}^\infty$ is a Cauchy sequence in k for each ν . Since k is complete, let $a_\nu = \lim_{m \rightarrow \infty} a_{m,\nu} \in k$. Define $f = \sum a_\nu \xi^\nu$. We first show that $f \in T_n$. For any $\varepsilon > 0$, there exists N such that $|a_{k,\nu} - a_{j,\nu}| \leq |f_k - f_j| < \varepsilon/2$ for all k, j . Fix $k > N$ and let $j \rightarrow \infty$ to get $|a_{k,\nu} - a_\nu| \leq \varepsilon/2$ by continuity of the valuation on k . Since $f_k \in T_n$, there exists M such that for all ν with $|\nu| > M$, one has $|a_{k,\nu}| < \varepsilon/2$. Then,

$$|a_\nu| \leq \max\{|a_\nu - a_{k,\nu}|, |a_{k,\nu}|\} \leq \varepsilon/2 < \varepsilon$$

for all $|\nu| > M$. Because $\varepsilon > 0$ was arbitrary, we obtain $|a_\nu| \rightarrow 0$. Hence $f \in T_n$. Additionally, one has

$$\max_\nu |a_{k,\nu} - a_\nu| = |f_k - f| \leq \varepsilon/2 < \varepsilon$$

for all $k > N$. Thus $f_k \rightarrow f$ in T_n . □

Remark 3.3. (i) Note that $\max_\nu |a_{k,\nu} - a_\nu|$ exists because $a_{k,\nu} - a_\nu$ are the coefficients of $f_k - f \in T_n$. Accordingly, $|a_{k,\nu} - a_\nu| \rightarrow 0$ and the $|a_{k,\nu} - a_\nu| \geq \varepsilon$ for finitely many ν argument applies.

(ii) Observe that our argument on the existence of $\alpha, \beta \in k$ shows that the value group of T_n and k are equal.

As in the proof, we define

$$\begin{aligned} T_n^\circ &= \{f \in T_n : |f| \leq 1\}, \\ T_n^\vee &= \{f \in T_n : |f| < 1\}, \\ \widetilde{T}_n &= T_n^\circ / T_n^\vee. \end{aligned}$$

Proposition 3.4. $\widetilde{T}_n \cong \widetilde{k}[\xi_1, \dots, \xi_n]$ canonically.

Proof. Consider the reduction map $T_n^\circ \rightarrow \widetilde{k}[\xi_1, \dots, \xi_n]$, $f \mapsto \widetilde{f}$ as defined in the proof of the theorem. This map is clearly a k° -algebra homomorphism. By definition, $\widetilde{f} = 0$ if and only if each coefficient of f is in $k^\vee$, which is equivalent to $|f| < 1$. Thus, $\ker(\widetilde{}) = T_n^\vee$. Moreover, the map is surjective because any polynomial in $\widetilde{k}[\xi_1, \dots, \xi_n]$ can be lifted to a polynomial in $k^\circ[\xi_1, \dots, \xi_n] \subset T_n^\circ$ by choosing preimages of its coefficients under the natural projection $\pi : k^\circ \rightarrow \widetilde{k}$. The desired isomorphism $\widetilde{T}_n \cong \widetilde{k}[\xi_1, \dots, \xi_n]$ then follows immediately from the first isomorphism theorem. $\square$

From this result, many properties of T_n can be understood more easily by passing to the residue field.

The n -dimensional unit ball is defined by

$$B^n(k) := \{(x_1, \dots, x_n) \in k^n : \max_i |x_i| \leq 1\}.$$

We now show that each element $f = \sum_\nu a_\nu \xi^\nu \in T_n$ defines a map $B^n(\overline{k}) \rightarrow \overline{k}$, denoted by f again. Let $x = (x_1, \dots, x_n) \in B^n(\overline{k})$ and $L \subset \overline{k}$ be a finite extension of k containing the coordinates $x_1, \dots, x_n$ of x (e.g., take $L = k(x_1, \dots, x_n)$). Then L is complete (see 3.2.4/2, [BGR84]). Since $|a_\nu| \rightarrow 0$ in k and the valuation on L extends that of k , we also have $|a_\nu| \rightarrow 0$ in L . This means that $\sum_\nu a_\nu x^\nu$ converges in L . Hence the function

$$f : B^n(\overline{k}) \rightarrow \overline{k}, \quad x \mapsto f(x) := \sum_\nu a_\nu x_1^{\nu_1} \dots x_n^{\nu_n}$$

is well-defined. In particular, $f(B^n(k)) \subset k$.

Conversely, suppose that $f : B^n(\overline{k}) \rightarrow \overline{k}$ admits a convergent power series expansion on $B^n(\overline{k})$ with coefficients in $\overline{k}$ with the additional property that $f(B^n(k)) \subset k$. Then the first condition implies that $f \in \overline{k}[[\xi_1, \dots, \xi_n]]$, i.e., $f = \sum_\nu a_\nu \xi^\nu$ with $a_\nu \in \overline{k}$. Since

$(1, 1, \dots, 1) \in B^n(\bar{k})$, we see that $|a_\nu| \rightarrow 0$. Next, write $f = f_1 + f_2$, where the nonzero coefficients of f_1 are not in k and those of f_2 are in k . Then $f_1(B^n(k)) \subset k$ because we have $f(B^n(k)) \subset k$ by assumption and $f_2(B^n(k)) \subset k$ by construction. Now suppose that $f_2 = 0$, so that every nonzero coefficient of f lies outside k . Also suppose that $f \neq 0$. Then we can write $f = \xi_n^s \sum_{\nu=s}^{\infty} g_\nu(\xi_1, \dots, \xi_{n-1}) \xi_n^{\nu-s}$ for some $s \geq 0$ and $g_\nu \in T_{n-1}(\bar{k})$ with $g_s \neq 0$. Clearly, the nonzero coefficients of g_s cannot be in k . Letting $(x_1, \dots, x_{n-1}) \in B^{n-1}(k)$, we see that $g_s(x_1, \dots, x_{n-1}) = \lim_{x_n \rightarrow 0} f(x_1, \dots, x_n) x_n^{-s} \in k$ as $f(B^n(k)) \subset k$. This shows that g_s has the same properties as f . Applying this argument n times, we get a nonzero constant in k that is a coefficient of f . This contradicts our assumption that f does not contain nonzero coefficients in k . This shows that $f = 0$. We conclude that if $f \in T_n(\bar{k})$ maps $B^n(k)$ into k , then $f \in T_n$.

In summary, we have proved the following result:

Proposition 3.5. The elements of $T_n(k)$ induce functions $f : B^n(\bar{k}) \rightarrow \bar{k}$ such that f admits a power series expansion with coefficients in $\bar{k}$ that converges on the unit ball $B^n(\bar{k})$ and maps $B^n(k)$ into k .

Furthermore, if $L \subset \bar{k}$ is any finite extension of k , then $f(B^n(L)) \subset L$ for all $f \in T_n(k)$.

Remark 3.6. (i) The usual $a_\nu = \frac{1}{\nu!} \frac{\partial^\nu f}{\partial x^\nu}(0) \in k$ argument we are familiar with cannot be employed here since k may have characteristic $p \neq 0$.

(ii) Similar to the relation between polynomials and polynomial functions, one can show that $f, g \in T_n$ give rise to the same function if and only if $f = g$.

Proposition 3.7. Let $f \in T_n$. Then $\sup_{x \in B^n(\bar{k})} |f(x)| \leq |f|$, and f induces a continuous function on $B^n(\bar{k})$.

Proof. Let $f = \sum_\nu a_\nu \xi^\nu$ and let $x \in B^n(\bar{k})$. Then $|x^\nu| \leq 1$ and we get

$$|f(x)| = \left| \sum_\nu a_\nu x^\nu \right| \leq \max_\nu |a_\nu x^\nu| \leq \max_\nu |a_\nu| = |f|,$$

where we used the ultrametric inequality in the first inequality. Taking sup over $x \in B^n(\bar{k})$ gives the first statement. To see that f is continuous, set $P_N(x) = \sum_{|\nu| \leq N} a_\nu x^\nu$. Since P_N is just a finite sum of monomials, it is clearly a continuous function on $B^n(\bar{k})$. One

has

$$|f(x) - P_N(x)| = \left| \sum_{|\nu| > N} a_\nu x^\nu \right| \leq \max_{|\nu| > N} |a_\nu x^\nu| \leq \max_{|\nu| > N} |a_\nu|.$$

Observe that the bound $\max_{|\nu| > N} |a_\nu|$ for $|f(x) - P_N(x)|$ does not depend on x . Since $|a_\nu| \rightarrow 0$ as $|\nu| \rightarrow \infty$, we see that P_N converges to f uniformly. Because the uniform limit of a sequence of continuous functions is continuous (this result has nothing to do with the field being non-archimedean or possibly having a positive characteristic), f is continuous on $B^n(\bar{k})$. $\square$

Let $\sim : k^\circ \rightarrow \bar{k}$ be the residue map and denote the map $B^n(\bar{k}) = \bar{k}^{\circ n} \rightarrow \bar{k}^{\sim n}$ given by $(x_1, \dots, x_n) \mapsto (\tilde{x}_1, \dots, \tilde{x}_n)$, where each $\tilde{x}_i$ is the residue of x_i in $\bar{k}$, by $\sim$ again. Then the diagram

$$\begin{array}{ccc} \bar{k}^{\circ n} & \xrightarrow{f} & \bar{k}^{\circ} \\ \sim \downarrow & & \downarrow \sim \\ \bar{k}^{\sim n} & \xrightarrow{\tilde{f}} & \bar{k}^{\sim} \end{array}$$

commutes for all $f \in T_n$, where $\tilde{f}$ is the map induced by $\tilde{f} \in \bar{k}^{\sim}[z_1, \dots, z_n]$. Letting $f = \sum_\nu a_\nu \xi^\nu$, one has

$$\widetilde{f(x)} = \widetilde{\left(\sum_\nu a_\nu x^\nu \right)} = \sum_\nu \widetilde{a_\nu x^\nu} = \sum_\nu \tilde{a}_\nu \tilde{x}^\nu$$

On the other hand, by the definition of the map induced by the polynomial $\tilde{f} \in \bar{k}^{\sim}[\xi_1, \dots, \xi_n]$, we have $\tilde{f}(x) = \sum_\nu \tilde{a}_\nu x^\nu$. Then $\tilde{f}(\tilde{x}) = \sum_\nu \tilde{a}_\nu \tilde{x}^\nu$. This confirms that the diagram is commutative.

We now use the above diagram to show that the inequality in Proposition (3.7) is actually an equality, which gives us an analog of the maximum modulus principle.

Theorem 3.8 (Maximum Modulus Principle). For each $f \in T_n$, there exists a point $x \in B^n(\bar{k})$ such that $|f(x)| = |f|$. If $|f(x)| = |f|$ for some point $x \in \bar{k}^{\vee n}$, then $|f(x)| = |f|$ for all $x \in \bar{k}^{\vee n}$. Moreover, the same results hold if $\bar{k}$ is replaced by any extension $L \subset \bar{k}$ of k , provided the residue field $\tilde{L}$ is infinite.

Proof. Similar to the proof of Theorem (3.2), we can find $c \in \bar{k}$ such that $|c| = |f|^{-1}$. Then defining $g = cf$ gives $|g| = 1$. If $|g(x)| = |g| = 1$, then $|cf(x)| = |f|^{-1}|f(x)| = 1$

shows that $|f(x)| = |f|$. So we can assume that $|f| = 1$. Because $\widetilde{\bar{k}} = \widetilde{\bar{k}}$ (see 3.4.1/4, [BGR84]), $\widetilde{\bar{k}}$ has infinitely many elements since it is algebraically closed. This forces the existence of a point $x = (x_1, \dots, x_n) \in B^n(\bar{k})$ such that $\widetilde{f}(\widetilde{x_1}, \dots, \widetilde{x_n}) \neq 0$. By the commutativity of the diagram above, this means that $\widetilde{f(x)} \neq 0$. Hence $|f(x)|_{\widetilde{\bar{k}}} = 1$.

To see the second statement, suppose $|f(x)| = |f| = 1$ for some $x \in \widetilde{\bar{k}}$. Observe that $\tilde{x} = (0, \dots, 0)$ for each $x \in \widetilde{\bar{k}}$. Then $\widetilde{f(x)} = \widetilde{f(\tilde{x})} = \widetilde{f(0, \dots, 0)} \neq 0$ since $|f| = 1$. Because $\widetilde{f(0, \dots, 0)}$ is the constant term of the reduced polynomial $\widetilde{f}$, call it $\widetilde{a_0}$, we see that $|\widetilde{a_0}| = 1$. Now write $f(y) = a_0 + \sum_{\nu \neq 0} a_\nu y^\nu$ for $y \in \widetilde{\bar{k}}$. Since $|y_i| < 1$ for each component y_i of y and $a_\nu \leq |f| = 1$ for all ν , we get $|\sum_{\nu \neq 0} a_\nu y^\nu| < 1$ by ultrametric inequality. Then

$$|f(y)| = \max \left(|a_0|, \left| \sum_{\nu \neq 0} a_\nu y^\nu \right| \right) = |a_0| = 1.$$

Finally, note that for the existence of x with $\widetilde{f(\tilde{x_1}, \dots, \tilde{x_n})} \neq 0$, we only used the fact that $\widetilde{\bar{k}}$ is infinite. So the same arguments apply if $\widetilde{L}$ is infinite. $\square$

Moreover, we have the following corollary:

Corollary 3.9. The maximum of $f \in T_n$ is attained on the set

$$S = \{(x_1, \dots, x_n) \in B^n(\bar{k}) : |x_1| = \dots = |x_n| = 1\}.$$

Proof. Consider the auxiliary series $g \in T_n$ defined by

$$g(\xi_1, \dots, \xi_n) = \xi_1 \xi_2 \dots \xi_n f(\xi_1, \dots, \xi_n).$$

First, we determine the Gauss norm of g . Recall that the Gauss norm is a valuation on T_n . Since each coordinate ξ_i is a polynomial with a leading coefficient of 1, we have $|\xi_i| = 1$. Then

$$|g| = |\xi_1| \cdot |\xi_2| \dots |\xi_n| \cdot |f| = |f|.$$

Applying Theorem (3.8) to g , there exists $x = (x_1, \dots, x_n) \in B^n(\bar{k})$ such that $|g(x)| = |g|$. Substituting our previous calculation, we obtain

$$|x_1| \cdot |x_2| \dots |x_n| \cdot |f(x)| = |f|.$$

By Proposition (3.7), $|f(x)| \leq |f|$. If $|x_i| < 1$ for some i , then $|x_1| \dots |x_n| |f(x)| < |f|$. This would contradict the equality $|g(x)| = |f|$. Thus, we must have $|x_i| = 1$ for all i , implying $|f(x)| = |f|$. This shows that $x \in S$, and that the maximum value $|f|$ is assumed at this point. $\square$

We now prove Remark (3.6):

Corollary 3.10 (Identity Theorem). If $f, f' \in T_n$ induce the same function on $B^n(\bar{k})$, then $f = f'$.

Proof. It suffices to show that $h = f - f' = 0$. We are given that h induces the zero function on $B^n(\bar{k})$. Then $h(x) = 0$ for all $x \in B^n(\bar{k})$, i.e., $0 = \sup_{x \in B^n(\bar{k})} |h(x)| = |h|$. This shows that $h = 0$ as $||$ is a norm. $\square$

For later references, we list some algebraic properties of the Tate algebra. These results all rely on analogs of Weierstrass Preparation and Division Theorems for the non-archimedean setting.

Proposition 3.11. We have the following properties:

- (i) T_n is Noetherian and a unique factorization domain.
- (ii) T_n is Jacobson, i.e., every prime ideal of T_n is the intersection of maximal ideals containing it.
- (iii) Each ideal of T_n is closed.
- (iv) For each maximal ideal $\mathfrak{m} \subset T_n$, the residue field $T_n/\mathfrak{m}$ is a finite extension of k .

Proof. See [BGR84]. $\square$

3.2 Maximal Ideal Spectrum of Tate Algebras

We now start relating our discussion on Tate algebras to geometry. We start with an observation that will help us in that goal, whose proof is straightforward:

Proposition 3.12. Let k be a valued field and X be a non-empty set. We denote the set of all bounded functions from X to k as $\mathcal{B}(X, k)$. Under the usual pointwise operations, $\mathcal{B}(X, k)$ becomes a k -algebra.

By this proposition, one can define a k -algebra semi-norm on $\mathcal{B}(X, k)$ by

$$|f| := \sup_{x \in X} |f(x)|.$$

In general, one tries to find a space X such that the above semi-norm becomes a k -algebra norm on $\mathcal{B}(X, k)$ and such that a k -algebra A may be identified as $\mathcal{B}(X, \bar{k})$. In our case, we will try to find X from $A = T_n$. First, a definition is needed:

Definition 3.13. Let A be a k -algebra. The set

$$\text{Max}_k A = \{x \subset A : x \text{ is a maximal ideal in } A \text{ and } A/x \text{ is an algebraic extension of } k\}.$$

is called the spectrum of maximal ideals of A .

Remark 3.14. Note that for T_n , Proposition (3.11) implies that $T_n/\mathfrak{m}$ is a finite extension of k and thus algebraic. So we can write $\text{Max } T_n$ instead of $\text{Max}_k T_n$.

For a monic polynomial $p(x) \in A[x]$ given by $p(x) = x^m + a_1x^{m-1} + \cdots + a_m$, we define its *spectral value* by

$$\sigma(p) = \max_{1 \leq i \leq m} |a_i|^{1/i}.$$

Then for each $y \in A/\mathfrak{m}_x$ ⁵, we define its *spectral norm* by

$$|y|_{sp} = \sigma(p),$$

where $p \in k[x]$ is the minimal polynomial of y over k (keep in mind that A/x is an algebraic extension of k). One can show that $||_{sp}$ is a power-multiplicative k -algebra norm on A/x extending the norm on k (see 3.2.1/2, [BGR84]).

Let $x \in \text{Max } A$ and let $f \in A$. We denote the image of f under the canonical projection $\pi_x : A \rightarrow A/x$ by $f(x)$. By the previous paragraph, one can give A/x the spectral norm. Dropping the subscript sp , we write $|f(x)|$ for the spectral norm of $f(x) \in A/x$. One is then led to the following definition:

⁵From now on, we will use x and $\mathfrak{m}_x$ to denote maximal ideals. Ideally, x is used when we think of the maximal ideal as a point in $\text{Max } A$ and $\mathfrak{m}_x$ is used when the maximal ideal is considered as an ideal of A . However, we do not pay attention to this distinction and use both notations interchangeably, trusting that the context makes the usage clear.

Definition 3.15. The supremum semi-norm $\| \cdot \|_{\text{sup}}$ on $\text{Max } A$ is given by

$$|f|_{\text{sup}} = \begin{cases} 0, & \text{if } \text{Max } A = \emptyset \\ \sup\{|f(x)| : x \in \text{Max } A\}, & \text{if } \text{Max } A \neq \emptyset \text{ and } f(\text{Max } A) \text{ is bounded} \\ \infty, & \text{otherwise} \end{cases}$$

We say that $\| \cdot \|_{\text{sup}}$ is finite if $f(\text{Max } A)$ is bounded for all $f \in A$.

We now show that T_n can be seen as an algebra of functions on $\text{Max } T_n$. Given $\mathfrak{m} \in \text{Max } A$, we have $f(\mathfrak{m}) \in T_n/\mathfrak{m}$. Because $T_n/\mathfrak{m}$ is a finite extension, we can embed it into $\bar{k}$. Consequently, $f(\mathfrak{m})$ can be considered as an element of $\bar{k}$. The issue here is that there is no canonical way to embed $T_n/\mathfrak{m}$ into $\bar{k}$, which means that $f(\mathfrak{m})$ does not directly correspond to a point, but rather to a set of conjugate points of $B^n(\bar{k})$ under the action of the Galois group $\text{Gal}(\bar{k}/k)$.

For $x \in B^n(\bar{k})$, set

$$\mathfrak{m}_x = \{f \in T_n : f(x) = 0\}.$$

Let $f = \sum_{\nu} a_{\nu} z^{\nu} \in T_n$. Pick $x = (x_1, \dots, x_n) \in B^n(\bar{k})$. Then $k(x) := k(x_1, \dots, x_n)$ is a finite extension of k , and we have $S_N = \sum_{|\nu| \leq N} a_{\nu} x^{\nu} \in k(x)$ for all N . The valuation on k extends uniquely to $k(x)$ such that $k(x)$ is complete (see [BGR84]). So $k(x)$ is a complete metric space and thus closed. By Proposition (3.5), the series $f(x)$ converges in $\bar{k}$. On the other hand, we have $\{S_n\}_{n \in \mathbb{N}} \subset k(x)$ and $S_n \rightarrow f(x)$. Because $k(x)$ is closed, we see that $f(x) \in k(x)$. It is thus possible to define a map

$$h_x : T_n \rightarrow k(x), \quad f \mapsto f(x)$$

with $\ker h_x = \mathfrak{m}_x$ by definition. Note that the coordinate functions $\xi_i \in T_n$ map to the coordinates of x , i.e., $h_x(\xi_i) = x_i$ for each i . Since h_x is a k -algebra homomorphism, $\text{im}(h_x)$ must contain the smallest k -subalgebra of $\bar{k}$ that contains $x_1, \dots, x_n$. This is the polynomial ring $k[x_1, \dots, x_n]$. Because $x \in \bar{k}$, each x_i is algebraic over k . Recall that a k -subalgebra of an algebraic extension that is an integral domain is a field. Since $k[x_1, \dots, x_n]$ is a k -algebra generated by algebraic elements, it is a field. Specifically,

$k[x_1, \dots, x_n] = k(x_1, \dots, x_n)$. Thus, we have

$$k(x) \subseteq \text{im}(h_x) \subseteq k(x),$$

which implies $\text{im}(h_x) = k(x)$. Then $T_n / \ker h_x \cong k(x)$ shows that $\mathfrak{m}_x = \ker h_x$ is a maximal ideal in T_n . We thus obtain a map

$$\tau : B^n(\bar{k}) \rightarrow \text{Max } T_n, \quad x \mapsto \mathfrak{m}_x.$$

Proposition 3.16. The diagram

$$\begin{array}{ccc} B^n(\bar{k}) & \xrightarrow{f} & \bar{k} \\ \tau \downarrow & & \downarrow \\ \text{Max } T_n & \xrightarrow{f} & \bar{k}/\Gamma \end{array}$$

commutes for all $f \in T_n$.

Proof. Let $x \in B^n(\bar{k})$. We saw that $h_x : T_n(k) \rightarrow k(x) \subset \bar{k}$ has kernel $\mathfrak{m}_x$. By the First Isomorphism Theorem, h_x factors through the residue field $T_n/\mathfrak{m}_x$. This yields a specific k -algebra embedding $i : T_n/\mathfrak{m}_x \hookrightarrow \bar{k}$ such that

$$f(x) = h_x(f) = i(f(\mathfrak{m}_x)).$$

Let $\pi : \bar{k} \rightarrow \bar{k}/\Gamma$ be the canonical projection mapping an element to its Galois orbit. Then

$$\pi(f(x)) = \pi(i(f(\mathfrak{m}_x))).$$

By definition, the value of f on $\text{Max } T_n$ at $\mathfrak{m}_x$ is exactly the Galois orbit of all possible embeddings of $f(\mathfrak{m}_x)$ into $\bar{k}$. Since i is one such embedding, $\pi(i(f(\mathfrak{m}_x)))$ is exactly this orbit. Hence, the diagram commutes. $\square$

Our discussion on maximal ideals of T_n culminates in the following theorem:

Theorem 3.17. The map $\tau : B^n(\bar{k}) \rightarrow \text{Max } T_n$ is surjective and has finite fibers. Furthermore, for any two points $x, y \in B^n(\bar{k})$, we have $\tau(x) = \tau(y)$ if and only if there exists

$\gamma \in \Gamma = \text{Gal}(\bar{k}/k)$ such that $\gamma(x) = y$. Consequently, τ induces a bijection

$$B^n(\bar{k})/\Gamma \xrightarrow{\sim} \text{Max } T_n.$$

In particular, if k is algebraically closed, then τ is a bijection.

Proof. We first demonstrate the surjectivity of τ . Let $\mathfrak{m} \in \text{Max } T_n$ be an arbitrary maximal ideal. Then $L = T_n/\mathfrak{m}$ is a finite (algebraic) extension of k . We can therefore choose a k -algebra embedding $i : L \hookrightarrow \bar{k}$. The composition $T_n \xrightarrow{\pi} L \xrightarrow{i} \bar{k}$ yields a continuous k -algebra homomorphism $\phi : T_n \rightarrow \bar{k}$ whose kernel is $\mathfrak{m}$. For each $j = 1, \dots, n$, let $x_j = \phi(\xi_j)$ be the image of the indeterminate ξ_j . Because ϕ is continuous, it is uniquely determined by the x_i . Thus, ϕ coincides with the evaluation map $h_x : T_n \rightarrow \bar{k}$ at the point $x = (x_1, \dots, x_n)$. Since $\ker(h_x) = \mathfrak{m}_x$, we conclude that $\mathfrak{m}_x = \mathfrak{m}$, proving τ is surjective.

Next, we examine the fibers of τ . Since τ is surjective, for a fixed $\mathfrak{m} \in \text{Max } T_n$, the points in $\tau^{-1}(\mathfrak{m})$ correspond bijectively to the distinct k -algebra embeddings of $T_n/\mathfrak{m}$ into $\bar{k}$. Because $T_n/\mathfrak{m}$ is a finite extension of k , there are only finitely many such embeddings. Consequently, the fibers of τ are strictly finite.

Finally, suppose we have $x, y \in B^n(\bar{k})$ with $\tau(x) = \tau(y) = \mathfrak{m}$. This implies that the evaluation maps h_x and h_y have the identical kernel $\mathfrak{m}$, which induces a k -isomorphism $k(x) \cong T_n(k)/\mathfrak{m} \cong k(y)$. This isomorphism explicitly maps $x_j \mapsto y_j$ for all j . Any such k -isomorphism between finite subextensions of $\bar{k}$ extends to an automorphism of $\bar{k}$, meaning there exists some $\gamma \in \Gamma$ such that $\gamma(x) = y$.

Conversely, if $\gamma(x) = y$, then for any $f \in T_n$, we have $f(y) = f(\gamma(x)) = \gamma(f(x))$. Since γ is an automorphism, $f(y) = 0$ if and only if $f(x) = 0$, which forces $\mathfrak{m}_x = \mathfrak{m}_y$. This establishes the orbit equivalence, and the bijection follows. $\square$

As a consequence of the preceding theorem, every maximal ideal $\mathfrak{m} \in \text{Max } T_n$ is of the form $\mathfrak{m}_x$ for some $x \in B^n(\bar{k})$. This justifies treating the maximal ideal spectrum as our fundamental geometric space. In other words, viewing the elements of the Tate algebra T_n as functions on the polydisc $B^n(\bar{k})$ or as functions on $\text{Max } T_n$ is essentially equivalent. More precisely, for any $f \in T_n(k)$, the induced function on the spectrum

$$f : \text{Max } T_n \rightarrow \bar{k}/\Gamma$$

is obtained from the standard evaluation map on $B^n(\bar{k})$ by taking the quotient via Γ (i.e., identifying Galois conjugate points). Moreover, the Identity Theorem implies that T_n can be identified with the induced algebra of functions on $\text{Max } T_n$.

3.3 Affinoid Algebras

In this subsection, all homomorphisms are understood as k -algebra homomorphisms.

Definition 3.18. A k -Banach algebra A is called k -affinoid if there exists a continuous surjective homomorphism $\alpha : T_n \rightarrow A$ for some $n \geq 0$.

From functional analysis, the quotient $T_n / \ker \alpha$ equipped with the norm

$$|\bar{f}| := \inf\{|h| : h \in \bar{f}\}$$

becomes a Banach space since $\ker \alpha$ is a closed subspace, where $\bar{f}$ denotes the image of $f \in T_n$ under the quotient map $\pi : T_n \rightarrow T_n / \ker \alpha$. Furthermore, the topology induced by this residue norm is the same as the quotient topology on $T_n / \ker \alpha$. To see this, first observe that taking $h = 0$ in $|\pi(f)| = \inf_{h \in \ker \alpha} |f + h|$ yields $|\pi(f)| \leq |f|$. So π is bounded and thus continuous. This means that every set open in the residue norm is also open in the quotient topology since the latter is the finest topology that makes the projection continuous. Now, consider the open unit ball $B_1(0, T_n) \subset T_n$. Since $|\pi(f)| \leq |f|$, we have $\pi(B_1(0, T_n)) \subset B_1(0, T_n / \ker \alpha) \subset T_n / \ker \alpha$. Conversely, let $\bar{f} \in T_n / \ker \alpha$ such that $|\bar{f}| < 1$. Then there exists $h \in \ker \alpha$ such that $|f + h| < 1$. Let $g = f + h$. Then $g \in B_1(0, T_n)$ and $\pi(g) = \bar{f}$. Hence $\pi(B_1(0, T_n)) = B_1(0, T_n / \ker \alpha)$. Because π is a linear operator, it commutes with translation and scalar multiplication. Therefore, mapping the open unit ball to the open unit ball guarantees that π maps every open ball to an open ball, making it an open mapping. Then for any open set U in the quotient topology, we have that $\pi(\pi^{-1}(U)) = U$ is open with respect to the residue norm. This shows that the topology induced by the residue norm is the same as the quotient topology on $T_n / \ker \alpha$.

Next, observe that

$$|\overline{fg}| = \inf_{h \in \ker \alpha} |fg + h| \leq \inf_{h_1, h_2 \in \ker \alpha} |(f + h_1)(g + h_2)| = \inf_{h_1, h_2 \in \ker \alpha} |f + h_1| |g + h_2| = |\bar{f}| |\bar{g}|,$$

where the inequality holds since $\{(f + h_1)(g + h_2) \mid h_1, h_2 \in I\} \subseteq \{fg + h \mid h \in I\}$ and tak-

ing $\inf$ reverses the inclusion. This shows that $T_n/\ker \alpha$ is a k -Banach algebra. Moreover, we claim that $T_n/\ker \alpha \cong A$ as k -Banach algebras. To see this, let $\bar{\alpha} : T_n/\ker \alpha \rightarrow A$ be given by $\bar{\alpha}(\bar{f}) = \alpha(f)$. Because α is continuous, it is bounded. So $\exists M$ such that $|\alpha(g)| \leq M|g|$ for all $g \in T_n$. Note that $|\bar{\alpha}(\bar{f})| = |\alpha(f+h)|$ for all $h \in \ker \alpha$ since $\alpha(f+h) = \alpha(f) + \alpha(h) = \alpha(f) + 0 = \alpha(f)$. Then $|\bar{\alpha}(\bar{f})| = |\alpha(f+h)| \leq M|f+h|$ for all $h \in \ker \alpha$ and taking $\inf$ over $\ker \alpha$ gives

$$|\bar{\alpha}(\bar{f})| \leq \inf_{h \in \ker \alpha} M|f+h| = M|\bar{f}|.$$

So $\bar{\alpha}$ is bounded and thus continuous. It is surjective since α is surjective. So the Open Mapping Theorem shows that $\bar{\alpha}$ is open. If $\bar{\alpha}(\bar{f}) = 0$ then $\alpha(f) = 0$, i.e., $f \in \ker \alpha$. This means that $\bar{f} = 0$. Hence $\bar{\alpha}$ is bijective and thus a homeomorphism since it is open, proving the claim.

Proposition 3.19. Let A be a k -affinoid algebra. Then A is Noetherian and Jacobson with each ideal $\mathfrak{a} \subset A$ closed. Moreover, $A/\mathfrak{a}$ is also k -affinoid if one considers the residue norm.

Proof. Let $\alpha : T_n \rightarrow A$ be as in the above definition. Then $T_n/\ker \alpha \cong A$ shows that A is a Noetherian Jacobson ring since T_n is and quotients of a Noetherian (resp. Jacobson) ring is Noetherian (resp. Jacobson). Note that $\alpha^{-1}(\mathfrak{a})$ is a closed ideal in T_n since all ideals of T_n are closed. Then $\mathfrak{a}$ is closed in A by definition of quotient topology. Finally, $A/\mathfrak{a}$ is k -affinoid via the composition $T_n \xrightarrow{\alpha} A \rightarrow A/\mathfrak{a}$. $\square$

We also have the following result, whose proof is again reduced to T_n :

Proposition 3.20. Let A be a k -affinoid algebra. If $\mathfrak{m}$ is a maximal ideal of A , then the residue field $A/\mathfrak{m}_A$ is a finite extension of k .

Proof. Let $\pi : T_n \rightarrow A$ be an epimorphism recognizing A as an affinoid algebra. Let $\mathfrak{m}_A$ be a maximal ideal of A . Under the identification $A \cong T_n/\ker \pi$, the ideal $\mathfrak{m}_A$ corresponds to an ideal $\mathfrak{m}/\ker \pi$, where $\mathfrak{m}$ is some ideal in T_n that contains $\ker \pi$. Recall that the maximal ideals of $T_n/\ker \pi$ are in bijection with the maximal ideals of T_n that contain $\ker \pi$. So $\pi^{-1}(\mathfrak{m}_A) = \mathfrak{m}$ is a maximal ideal in the T_n . By the Third Isomorphism Theorem, we have

$$(T_n/\ker \pi)/(\mathfrak{m}/\ker \pi) \cong T_n/\mathfrak{m}.$$

Substituting our established notations back, we get $A/\mathfrak{m}_A \cong T_n/\mathfrak{m}$. Since $T_n/\mathfrak{m}$ is a finite extension of k , it follows that $A/\mathfrak{m}_A$ is also a finite extension of k . $\square$

Similar to $\text{Max } T_n$, we will write $\text{Max } A$ instead of $\text{Max}_k A$ for affinoid varieties.

3.4 Affinoid Varieties

Now that we view the Tate algebra T_n as a ring of functions on the spectrum $\text{Max } T_n$, we can study the geometric subsets defined by the vanishing of these functions.

Definition 3.21. Let $F \subset T_n(k)$. The zero sets of F defined by

$$\begin{aligned}\overline{V}(F) &:= \{x \in B^n(\overline{k}) : f(x) = 0 \text{ for all } f \in F\} \subset B^n(\overline{k}) \\ V(F) &:= \{\mathfrak{m} \in \text{Max } T_n : f(\mathfrak{m}) = 0 \text{ for all } f \in F\} \subset \text{Max } T_n\end{aligned}$$

are called affinoid subsets of $B^n(\overline{k})$ and $\text{Max } T_n$, respectively.

As in the case of $V(I)$ for schemes, if $\mathfrak{a} \subset T_n$ is the ideal generated by F , one has $V(F) = V(\mathfrak{a})$. Furthermore, because T_n is Noetherian, we can write $\mathfrak{a} = (f_1, \dots, f_r)$ for some $f_1, \dots, f_r \in T_n$. Then it is clear that $x \in \text{Max } T_n$ is in $V(\mathfrak{a})$ if and only if it is a zero of the generators f_i , i.e., $V(\mathfrak{a}) = V(f_1, \dots, f_r)$.

Given $E \subset \text{Max } T_n$, we set

$$\text{id}(E) := \{f \in T_n : f|_E = 0\} = \bigcap_{x \in E} \mathfrak{m}_x.$$

Theorem 3.22 (Nullstellensatz for T_n). For any ideal $\mathfrak{a} \subset T_n$, we have

$$\text{id}(V(\mathfrak{a})) = \text{rad}(\mathfrak{a}).$$

Proof. By definition, $V(\mathfrak{a})$ consists of all maximal ideals $\mathfrak{m} \in \text{Max } T_n$ that contain $\mathfrak{a}$. The ideal $\text{id}(V(\mathfrak{a}))$ is the intersection of all maximal ideals in $V(\mathfrak{a})$, which means it is the intersection of all maximal ideals containing $\mathfrak{a}$. Because T_n is Jacobson, we see that $\text{id}(V(\mathfrak{a})) = \text{rad}(\mathfrak{a})$. $\square$

Proposition 3.23. The maps V and id satisfy the following properties:

(i) $V(\mathfrak{a}) \supset V(\mathfrak{b})$ for $\mathfrak{a} \subset \mathfrak{b} \subset T_n$.

- (ii) $\text{id}(E) \supset \text{id}(E')$ for $E \subset E' \subset \text{Max } T_n$.
- (iii) $V(\sum_{i \in I} \mathfrak{a}_i) = \bigcap_{i \in I} V(\mathfrak{a}_i)$ for a family $(\mathfrak{a}_i)_{i \in I}$ of ideals in T_n .
- (iv) $V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b})$ for ideals $\mathfrak{a}, \mathfrak{b} \subset T_n$.

Proof. The first three properties are simple. For (iv), note that $\mathfrak{a}\mathfrak{b} \subset \mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{a}, \mathfrak{b}$ gives $V(\mathfrak{a}) \cup V(\mathfrak{b}) \subset V(\mathfrak{a} \cap \mathfrak{b}) \subset V(\mathfrak{a}\mathfrak{b})$ by (i). Conversely, if $x \notin V(\mathfrak{a}) \cup V(\mathfrak{b})$ then there exist $f \in \mathfrak{a}$ and $g \in \mathfrak{b}$ with $f(x) \neq 0$ and $g(x) \neq 0$. Thus $(fg)(x) \neq 0$. Then $x \notin V(\mathfrak{a}\mathfrak{b})$ since $fg \in \mathfrak{a}\mathfrak{b}$. This shows that $V(\mathfrak{a}\mathfrak{b}) \subset V(\mathfrak{a}) \cup V(\mathfrak{b})$ and finishes the proof. $\square$

By this proposition, we see that the affinoid subsets of $\text{Max } T_n$ form the closed subsets of a topology on $\text{Max } T_n$. We call this topology the Zariski topology on $\text{Max } T_n$. One can show fairly easily that many of the results and concepts for affine varieties, such as irreducibility and finite decomposition of reducible varieties, are also valid for this topology.

We have seen that an element of T_n can be seen as a function on $\text{Max } T_n$ and $B^n(\bar{k})$. One can carry out a similar procedure for any affinoid subset. More concretely, let $\mathfrak{a} \subset T_n$ be an ideal and let $X = V(\mathfrak{a}) \subset \text{Max } T_n$ be the corresponding affinoid subset. Then an element of T_n gives a function on X simply by restricting the domain $\text{Max } T_n$ to X . Similar to the case of affine varieties, $f, g \in T_n$ determine the same function on X if and only if $f - g \in \text{id}(X)$, i.e., $f \equiv g \pmod{\text{id}(X)}$.

We can also see the elements of $A = T_n/\mathfrak{a}$ as functions on $\text{Max } A$. The idea is the same as in subsection 3.2. Namely, given $f \in A$ and $x \in \text{Max } A$, define $f(x)$ as the residue class of f in $A/\mathfrak{m}_x$. We can also define V and id in the same way. For $F \subset A$ and $E \subset \text{Max } A$, put

$$V(F) = \{x \in \text{Max } A : f(x) = 0 \text{ for all } f \in F\},$$

$$\text{id}(E) = \{f \in A : f(x) = 0 \text{ for all } x \in E\}.$$

The set $V(F)$ is called an affinoid subset of $\text{Max } A$. All the properties satisfied in the case of $A = T_n$, such as the Nullstellensatz, are satisfied for an arbitrary A because the properties of T_n (Jacobson and Noetherian) used to prove those results hold for A as well.

Let $\sigma : T_n \rightarrow A$ be the canonical projection, which induces a map

$${}^a\sigma : \text{Max } A \rightarrow \text{Max } T_n, \quad x \mapsto \sigma^{-1}(\mathfrak{m}_x).$$

To see that this map is well-defined, consider the composition $T_n \xrightarrow{\sigma} A \rightarrow A/\mathfrak{m}_x$. The kernel of this composition is the set of elements $f \in T_n$ such that $\sigma(f) \in \mathfrak{m}_x$, which is $\sigma^{-1}(\mathfrak{m}_x)$. By the First Isomorphism Theorem, $T_n/\sigma^{-1}(\mathfrak{m}_x) \cong A/\mathfrak{m}_x$. Since the quotient is a field, the ideal $\sigma^{-1}(\mathfrak{m}_x)$ must be maximal. The map ${}^a\sigma$ is also injective. Suppose $x, y \in \text{Max } A$ such that ${}^a\sigma(x) = {}^a\sigma(y)$. Then $\sigma^{-1}(\mathfrak{m}_x) = \sigma^{-1}(\mathfrak{m}_y)$. Applying σ to both sides gives $\mathfrak{m}_x = \mathfrak{m}_y$ because σ is surjective. Moreover, ${}^a\sigma$ is bijective onto X by the correspondence between the maximal ideals of A and the maximal ideals of T_n containing $\ker \sigma = \mathfrak{a}$. Hence we can identify $\text{Max } A$ with X to see that the restriction of $f \in T_n$ to X can be identified with $\sigma(f)$ viewed as a function on $\text{Max } A$.

3.4.1 Affinoid Maps

Definition 3.24. Let $X \subset B^n(\bar{k})$ be an affinoid subset and let $\phi : X \rightarrow B^n(\bar{k})$ be a map. We say that ϕ is an affinoid map if there are functions $f_1, \dots, f_n \in T_m/\text{id}(X)$ with $\phi(x) = (f_1(x), \dots, f_n(x))$ for each $x \in X$.

The affinoid map $\phi : X \rightarrow B^n(\bar{k})$ gives rise to a unique continuous k -algebra homomorphism (see 6.1.1/4, [BGR84])

$$\phi^* : T_n \rightarrow T_m/\text{id}(X), \quad g \mapsto g \circ \phi.$$

It is clear that we can obtain ϕ from ϕ^* by taking $g \in T_n$ to be the coordinate functions $\xi_i \in T_n$. This gives $\phi^*(\xi_i)(x) = \xi_i \circ \phi(x) = f_i(x)$ for any $x \in X$, so that $f_i = \phi^*(\xi_i)$. Thus $\phi \mapsto \phi^*$ is a one-to-one correspondence between affinoid maps $X \rightarrow B^n(\bar{k})$ and k -algebra homomorphisms $T_n \rightarrow T_m/\text{id}(X)$ (for continuity of $T_n \rightarrow T_m/\text{id}(X)$, see 6.1.3/1, [BGR84]).

The correspondence mentioned in the last paragraph can be improved as follows. We first show that if $Y \subset B^n(\bar{k})$ is an affinoid subset, then $\phi(X) \subset Y$ if and only if $\text{id}(Y) \subset \ker \phi^*$. To see this, take $g \in \text{id}(Y)$. Then $g(y) = 0$ for all $y \in Y$. Since $\phi(X) \subset Y$, $g(\phi(x)) = 0$ for all $x \in X$. Hence $\phi^*(g) = 0$ and $g \in \ker \phi^*$. Conversely, suppose that $\text{id}(Y) \subset \ker \phi^*$. For any $g \in \text{id}(Y)$, $g(\phi(x)) = 0$ for all $x \in X$. Recalling $V(\text{id}(Y)) = Y$, we get $\phi(x) \in Y$. So $\phi(X) \subset Y$. Now let $\phi : X \rightarrow Y \subset B^n(\bar{k})$ be an affinoid map. Then $\text{id}(Y) \subset \ker \phi^*$ and the universal property of quotients applied to ϕ^* implies that there is a unique k -algebra homomorphism $\sigma : T_n/\text{id}(Y) \rightarrow T_m/\text{id}(X)$ such

that $\sigma(\bar{g}) = \phi^*(g)$, where $\bar{g}$ is the image of g under the canonical projection. Conversely, every k -algebra homomorphism $\sigma : T_n/\text{id}(Y) \rightarrow T_m/\text{id}(X)$ can be pre-composed with the quotient map $\pi : T_n \rightarrow T_n/\text{id}(Y)$ to give a map $\phi^* = \sigma \circ \pi$. Because $\text{id}(Y) \subset \ker \phi^*$, we have $\phi(X) \subset Y$. Hence we obtain a one-to-one correspondence between affinoid maps $X \rightarrow Y$ and k -algebra homomorphisms $T_n/\text{id}(Y) \rightarrow T_m/\text{id}(X)$.

Suppose $\sigma : B \rightarrow A$ is a k -algebra homomorphism, where $A = T_m/\text{id}(X)$ and $B = T_n/\text{id}(Y)$ are k -affinoid algebras for some affinoid subsets $X \subset B^m(\bar{k})$ and $Y \subset B^n(\bar{k})$. For $x \in \text{Max } A$, considering the composition $B \xrightarrow{\sigma} A \rightarrow A/\mathfrak{m}_x$ yields a map

$$B/\sigma^{-1}(\mathfrak{m}_x) \hookrightarrow A/\mathfrak{m}_x$$

as before. Because $A/\mathfrak{m}_x$ is a finite extension of k and $B/\sigma^{-1}(\mathfrak{m}_x)$ is a k -subalgebra of $A/\mathfrak{m}_x$, we see that $\sigma^{-1}(\mathfrak{m}_x)$ is maximal. Hence, one gets a map

$${}^a\sigma : \text{Max } A \rightarrow \text{Max } B, \quad x \mapsto \sigma^{-1}(\mathfrak{m}_x).$$

Note that $\text{id}(X)$ gives rise to an affinoid subset in $\text{Max } T_m$ obtained from X by identifying conjugate points that can be identified with $\text{Max } A$. Define $\text{id} : X \rightarrow \text{Max } A$ by sending $x \in X$ to the maximal ideal of all functions in A vanishing at x . Similarly, one has the map $\text{id}' : Y \rightarrow \text{Max } B$. Denote the affinoid map corresponding to $\sigma : B \rightarrow A$ by $\phi : X \rightarrow Y$. More explicitly, if $B \cong T_n/\mathfrak{a}$, set $\sigma(\bar{\xi}_i) =: f_i$ and define $\phi(x) = (f_1(x), \dots, f_n(x))$. Since the $\bar{\xi}_i$ generate B , we have $g = g(\bar{\xi}_1, \dots, \bar{\xi}_n)$ for each $g \in B$. Then $\sigma(g(\bar{\xi}_1, \dots, \bar{\xi}_n)) = g(\sigma(\bar{\xi}_1), \dots, \sigma(\bar{\xi}_n)) = g(f_1, \dots, f_n)$ as σ is a homomorphism. Then $\sigma(g)(x) = g(f_1(x), \dots, f_n(x)) = g \circ \phi(x)$. Thus the diagram

$$\begin{array}{ccc} X & \xrightarrow{\phi} & Y \\ \text{id} \downarrow & & \downarrow \text{id}' \\ \text{Max } A & \xrightarrow{{}^a\sigma} & \text{Max } B \end{array}$$

is commutative because

$$\text{id}'(\phi(x)) = \{f \in B : f(\phi(x)) = 0\}$$

and

$${}^a\sigma \circ \text{id}(x) = \sigma^{-1}(\mathfrak{m}_x) = \{f \in B : \sigma(f) \in \mathfrak{m}_x\} = \{f \in B : \sigma(f)(x) = 0\},$$

where $\text{id}(x) = \mathfrak{m}_x = \{h \in A : h(x) = 0\}$.

We can now give the definition of affinoid varieties and morphisms between them:

Definition 3.25. A k -affinoid variety is a pair $\text{Sp } A = (\text{Max } A, A)$ with A a k -affinoid algebra. A morphism $\phi : \text{Sp } A \rightarrow \text{Sp } B$ of k -affinoid varieties is a pair $\phi = ({}^a\sigma, \sigma)$, where $\sigma : B \rightarrow A$ is a k -algebra homomorphism and ${}^a\sigma : \text{Max } A \rightarrow \text{Max } B$ is the map obtained from σ .

Following standard convention, we will use $\text{Sp } A$ to denote both the affinoid variety and its underlying set of maximal ideals $\text{Max } A$. Similarly, we will denote the homomorphism ${}^a\sigma$ by ϕ .

We will need the following result, whose proof is somewhat categorical and thus omitted, when we discuss G -topologies:

Proposition 3.26. If $\phi : \text{Sp } B \rightarrow \text{Sp } A$ is an affinoid map and $\text{Sp } A' \subset \text{Sp } A$ is an affinoid subdomain of $\text{Sp } A$, then $\phi^{-1}(\text{Sp } A')$ is an affinoid subdomain of $\text{Sp } B$.

Proof. See 7.2.2/4, [BGR84]. □

3.5 The Canonical Topology on an Affinoid Variety

Consider the unit ball $B^n(\bar{k})$ with subspace topology induced by the product topology on $\bar{k}^n$. For $z = (z_1, \dots, z_n) \in \bar{k}^n$, the function defined by

$$|z| := \max_{1 \leq i \leq n} |z_i|$$

is seen to be a norm on $\bar{k}^n$, where the ultrametric inequality follows from

$$\max_{1 \leq i \leq n} |z_i + w_i| \leq \max_{1 \leq i \leq n} (\max(|z_i|, |w_i|)) = \max(\max_{1 \leq i \leq n} |z_i|, \max_{1 \leq i \leq n} |w_i|) = \max(|z|, |w|).$$

Since all norms on a finite-dimensional vector space over a complete field are equivalent, $||$ defines the topology of $\bar{k}^n$ and so that of $B^n(\bar{k})$.

Definition 3.27. Let $\tau : B^n(\bar{k}) \rightarrow \text{Max } T_n$ be the surjection in Theorem (3.17). The quotient topology on $\text{Max } T_n$ induced by τ is called the canonical topology on $\text{Max } T_n$ (or $\text{Sp } T_n$).

We now define a topology on $\text{Sp } A$ for any k -affinoid algebra A . Let $\sigma : T_n \rightarrow A$ be an epimorphism recognizing A as a k -affinoid algebra. Then using the injection ${}^a\sigma : \text{Sp } A \hookrightarrow \text{Sp } T_n$, $x \mapsto \sigma^{-1}(\mathfrak{m}_x)$, we give $\text{Sp } A$ the subspace topology inherited from $\text{Sp } T_n$ and call it the canonical topology on $\text{Sp } A$. More concretely, a subset $U \subseteq \text{Sp } A$ is open if there exists an open set $W \subseteq \text{Sp } T_n$ such that $U = ({}^a\sigma)^{-1}(W)$. It can be shown that subsets of the form

$$(\text{Sp } A)(f_1, \dots, f_r) := \{x \in \text{Sp } A : |f_1(x)| \leq 1, \dots, |f_r(x)| \leq 1\}$$

for $f_1, \dots, f_r \in A$ form a basis for the topology of $\text{Sp } A$ (see 7.2.1/4, [BGR84]). Hence, the definition of the canonical topology is independent of the epimorphism σ .

3.5.1 Affinoid Subdomains

In classical topology, an open subset of a space naturally inherits the subspace topology. In rigid analytic geometry, we want certain subsets to inherit not just a topology, but the full geometric structure of an affinoid variety. One might sense that such a requirement leads to a universal property. The following definition confirms that intuition:

Definition 3.28. Let $\text{Sp } A$ be an affinoid variety with $U \subset \text{Sp } A$. We say that a morphism $\phi : \text{Sp } A' \rightarrow \text{Sp } A$ represents all affinoid maps into U if $\phi(\text{Sp } A') \subset U$ and satisfies the following property:

Given a morphism $\psi : \text{Sp } B \rightarrow \text{Sp } A$ with $\psi(\text{Sp } B) \subset U$, there exists a unique morphism $\psi' : \text{Sp } B \rightarrow \text{Sp } A'$ such that the diagram

$$\begin{array}{ccc} & \text{Sp } B & \\ \psi' \swarrow & & \searrow \psi \\ \text{Sp } A' & \xrightarrow{\phi} & \text{Sp } A \end{array}$$

commutes.

The following proposition gives some useful properties of ϕ with the above universal

property:

Proposition 3.29. Let $U \subset \mathrm{Sp} A$ and let $\phi : \mathrm{Sp} A' \rightarrow \mathrm{Sp} A$ be a morphism that represents all affinoid maps into U . Then

- (i) ϕ is injective with $\phi(\mathrm{Sp} A') = U$.
- (ii) For $x \in \mathrm{Sp} A'$, the homomorphism $\sigma : A \rightarrow A'$ induces an isomorphism $A/\mathfrak{m}_{\phi(x)}^n \cong A'/\mathfrak{m}_x^n$ for any $n \in \mathbb{N}$.
- (iii) For $x \in \mathrm{Sp} A'$, the maximal ideal $\mathfrak{m}_x \subset A'$ is generated by $\sigma(\mathfrak{m}_{\phi(x)})$, that is, $\mathfrak{m}_x = \sigma(\mathfrak{m}_{\phi(x)})A'$.

Proof. See 7.2.2/1, [BGR84]. □

By this proposition, we can identify the underlying set of $\mathrm{Sp} A'$ with U , which equips U with the structure of an affinoid variety.

Definition 3.30. Let $\mathrm{Sp} A$ be an affinoid variety with $U \subset \mathrm{Sp} A$. We say U is an affinoid subdomain of $\mathrm{Sp} A$ if there exists a morphism $\phi : \mathrm{Sp} A' \rightarrow \mathrm{Sp} A$ representing all affinoid maps into U .

Affinoid subdomains are crucial to understanding how rigid analytic geometry differs from ordinary geometry. However, due to their technical and tedious nature, we content ourselves with citing some results that will be important later, but whose proofs are rather meticulous and omitted.

Theorem 3.31. The affinoid subdomains of $\mathrm{Sp} A$ form a basis for the canonical topology on $\mathrm{Sp} A$. In particular, affinoid subdomains are open, and the intersection of affinoid subdomains is an affinoid subdomain.

Proof. See 7.2.5/4, [BGR84] □

Proposition 3.32. Let $\phi : \mathrm{Sp} B \rightarrow \mathrm{Sp} A$ be an affinoid map. If $\mathrm{Sp} A' \subset \mathrm{Sp} A$ is an affinoid subdomain of $\mathrm{Sp} A$, then $\phi^{-1}(\mathrm{Sp} A')$ is an affinoid subdomain of $\mathrm{Sp} B$.

Proof. See 7.2.2/4, [BGR84] □

Let X be an affinoid variety. Then each affinoid subdomain of X is of the form $\mathrm{Sp} A$ by Proposition (3.29). For each affinoid subdomain $\mathrm{Sp} A \subset X$, set $\mathcal{O}_X(\mathrm{Sp} A) = A$. Also, for each inclusion $\phi : \mathrm{Sp} A' \hookrightarrow \mathrm{Sp} A$, set $\mathcal{O}_X(\phi) = \sigma$, where $\sigma : A' \rightarrow A$ is as

in the definition of an affinoid variety. Hence $\mathcal{O}_X$ is a presheaf on the basis of affinoid subdomains.

Now let $\mathcal{U} = \{U_i\}_{i \in I}$ be a finite covering (i.e., I is a finite set) of X by affinoid subdomains $U_i \subset X$, called an affinoid covering. The following result will be our analog for the structure sheaf of a rigid variety when we introduce G -topologies.

Proposition 3.33. Let $\mathcal{U} = \{U_i\}_{i \in I}$ be an affinoid covering of an affinoid variety X . Then

- (i) If $f, g \in \mathcal{O}_X(X)$ such that $f|_{U_i} = g|_{U_i}$ for all $i \in I$, then $f = g$.
- (ii) If $f_i \in \mathcal{O}_X(U_i)$ such that $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for all $i, j \in I$, then there exists a (unique) element $f \in \mathcal{O}_X(X)$ such that $f|_{U_i} = f_i$ for each $i \in I$.

Proof. We will use Tate's Acyclicity Theorem (see 8.2.1/1, [BGR84]). By this theorem, the sequence

$$0 \rightarrow \mathcal{O}_X(X) \xrightarrow{\varepsilon} C^0(\mathcal{U}, \mathcal{O}_X) \xrightarrow{d^0} C^1(\mathcal{U}, \mathcal{O}_X) \rightarrow \dots$$

is exact, where ε is the augmentation map and $d^0, d^1, \dots$ are the coboundary maps and $C^k(\mathcal{U}, \mathcal{O}_X) = \prod_{(i_0, \dots, i_k) \in I^{k+1}} \mathcal{O}_X(U_{i_0} \cap \dots \cap U_{i_k})$ for each $k \geq 0$. In particular, one has $\varepsilon(f) = (f|_{U_i})_{i \in I}$. Then exactness at $\mathcal{O}_X(X)$ implies that ε is injective, so that (i) holds. For (ii), first note that $C^0(\mathcal{U}, \mathcal{O}_X) = \prod_{i \in I} \mathcal{O}_X(U_i)$. Then by definition of the coboundary maps, we have $d^0((f_i))_{ij} = f_j|_{U_i \cap U_j} - f_i|_{U_i \cap U_j}$ for each $(f_i)_{i \in I} \in C^0(\mathcal{U}, \mathcal{O}_X)$. Exactness at $C^0(\mathcal{U}, \mathcal{O}_X)$ then means that for $(f_i)_{i \in I} \in \ker d^0$, there exists $f \in \mathcal{O}_X(X)$ such that $f|_{U_i} = f_i$. Uniqueness of f follows from (i) by a familiar argument from the definition of a sheaf. $\square$

3.6 G -Topologies

To construct a global geometric theory over non-archimedean fields, one cannot rely on the standard metric topology induced by the valuation, as it renders the space totally disconnected. Instead, one needs the notion of G -topologies, also known as Grothendieck topologies. In this subsection, we will briefly introduce the basic concepts of these topologies.

Let X be a set and let $U \subset X$. A covering of U means a family $\{U_i\}_{i \in I}$ of subsets of X such that $U = \bigcup_{i \in I} U_i$. For coverings $\mathcal{U} = \{U_i\}_{i \in I}$ and $\mathcal{V} = \{V_j\}_{j \in J}$ of U , we say that $\mathcal{V}$ refines $\mathcal{U}$ if there is a map $\alpha : J \rightarrow I$ with $V_j \subset U_{\alpha(j)}$ for each $j \in J$.

Definition 3.34. A G -topology $\mathfrak{T}$ on X consists of

- (a) a system $\mathcal{S}$ of subsets in X , called admissible open or $\mathfrak{T}$ -open subsets of X , and
- (b) a family $\{\text{Cov } U\}_{U \in \mathcal{S}}$ of systems of coverings, called admissible coverings or $\mathfrak{T}$ -coverings, where $\text{Cov } U$ for $U \in \mathcal{S}$ contains coverings $\{U_i\}_{i \in I}$ of U by sets $U_i \in \mathcal{S}$.

These are subject to the following conditions:

- (i) If $U, V \in \mathcal{S}$, then $U \cap V \in \mathcal{S}$.
- (ii) If $U \in \mathcal{S}$, then $\{U\} \in \text{Cov } U$.
- (iii) If $U \in \mathcal{S}$, $\{U_i\}_{i \in I} \in \text{Cov } U$, and $\{V_{ij}\}_{j \in J_i} \in \text{Cov } U_i$ for $i \in I$, then $\{V_{ij}\}_{i \in I, j \in J_i} \in \text{Cov } U$.
- (iv) If $U, V \in \mathcal{S}$ with $V \subseteq U$ and $\{U_i\}_{i \in I} \in \text{Cov } U$, then $\{V \cap U_i\}_{i \in I} \in \text{Cov } V$.

If the above conditions are satisfied, we call X a G -topological space and denote it by $X_{\mathfrak{T}}$.

There is a natural interplay between G -topologies and ordinary topologies. Given a G -topology $\mathfrak{T}$ on X , one can associate an ordinary topology by simply taking the system of all $\mathfrak{T}$ -open subsets as a basis (if X is not covered by its $\mathfrak{T}$ -open subsets, we also include X in the basis). Conversely, starting with an ordinary topological space X and fixing a basis $\mathcal{B}$ of open sets that is closed under finite intersections (i.e., $U, V \in \mathcal{B} \implies U \cap V \in \mathcal{B}$), one can define a G -topology. We take $\mathcal{S} := \mathcal{B}$ as the system of admissible open sets. For any $U \in \mathcal{S}$, we can define the admissible coverings, $\text{Cov } U$, as either the system of all coverings $\{U_i\}_{i \in I}$ with $U_i \in \mathcal{S}$, or as the system of all finite coverings. Choosing finite coverings generally leads to a different G -topology.

Ultimately, the concept of ordinary topologies is completely subsumed and generalized by that of G -topologies. In the context of rigid analytic geometry, the definition of a G -topology provides exactly the minimal amount of topological structure required so that sheaf theory can be developed at a functional level, bypassing the totally disconnected nature of the canonical topology.

Definition 3.35. A map $\phi : X \rightarrow Y$ between G -topological spaces is called continuous if $\phi^{-1}(U)$ is admissible open in X for any admissible open set $U \subset Y$ and the system

$\{\phi^{-1}(U_i)\}_{i \in I}$ is an admissible covering of $\phi^{-1}(U)$ for any admissible covering $\{U_i\}_{i \in I}$ of an admissible open set $U \subset Y$.

By this definition, the notions of weakness and fineness for two G -topologies on X can be defined as in ordinary topology. Namely, if $\mathfrak{T}$ and $\mathfrak{T}'$ are two G -topologies on X , then $X_{\mathfrak{T}'}$ is finer than $X_{\mathfrak{T}}$ if the identity map $X_{\mathfrak{T}'} \rightarrow X_{\mathfrak{T}}$ is continuous. This means that each $\mathfrak{T}$ -open subset of X is also $\mathfrak{T}'$ -open and each $\mathfrak{T}$ -covering is also a $\mathfrak{T}'$ -covering.

We will also need the following definition:

Definition 3.36. Let X be a set equipped with two G -topologies $\mathfrak{T}$ and $\mathfrak{T}'$. We say that $\mathfrak{T}'$ is slightly finer than $\mathfrak{T}$ if the following are met:

- (i) $\mathfrak{T}'$ is finer than $\mathfrak{T}$.
- (ii) The $\mathfrak{T}$ -open sets form a basis for $\mathfrak{T}'$.
- (iii) If U is a $\mathfrak{T}$ -open set with a $\mathfrak{T}'$ -covering $\mathcal{U}$, then there exists a $\mathfrak{T}$ -covering refining $\mathcal{U}$.

To ensure that G -topologies behave like a workable geometry, we impose the following additional axioms:

- (G_0) Both $\emptyset$ and X must be admissible open subsets of X .
- (G_1) Let $U \subset X$ be an admissible open set, and let $V \subset U$ be an arbitrary subset. Suppose there exists an admissible covering $\{U_i\}_{i \in I}$ of U such that the intersection $V \cap U_i$ is an admissible open set in X for every index $i \in I$. Then, V must be an admissible open subset of X .
- (G_2) Let $\mathcal{U} = \{U_i\}_{i \in I}$ be a covering of an admissible open set $U \subset X$, where each U_i is an admissible open set in X . Suppose also that $\mathcal{U}$ has a refinement that is admissible. Then $\mathcal{U}$ is an admissible covering of U .

Let $\mathcal{C}$ be a category whose objects are sets and suppose that each $X \in \mathcal{C}$ has a G -topology $\mathfrak{T}_X$, so that morphisms are continuous maps between the $\mathfrak{T}_X$. We denote the collection of all G -topologies $\mathfrak{T}_X$ by $\mathfrak{T}$ and call it a G -topology on $\mathcal{C}$. For G -topologies $\mathfrak{T}$ and $\mathfrak{T}'$ on $\mathcal{C}$, $\mathfrak{T}'$ is said to be (slightly) finer than $\mathfrak{T}$ if $\mathfrak{T}'_X$ is (slightly) finer than $\mathfrak{T}_X$ for each $X \in \mathcal{C}$.

The following result will be useful later, whose proof requires several lemmas and is thus omitted:

Proposition 3.37. For a G -topology $\mathfrak{T}$ on $\mathcal{C}$, there is a unique finest G -topology $\mathfrak{T}'$ on $\mathcal{C}$ among all the G -topologies slightly finer than $\mathfrak{T}$. This G -topology satisfies (G_1) and (G_2) . Moreover, it satisfies (G_0) if $\mathfrak{T}$ does.

Proof. See 9.1.2/2, [BGR84]. □

In the statement, finest means that any G -topology $\mathfrak{T}''$ that is slightly finer than $\mathfrak{T}$ is coarser than $\mathfrak{T}'$.

3.6.1 Topology of Affinoid Varieties Revisited

Recall that the intersection of two affinoid subdomains of an affinoid variety X is also an affinoid subdomain of X . This suggests that we can introduce a G -topology on X . This is simply done by declaring all affinoid subdomains of X to be admissible open and all affinoid coverings of affinoid domains admissible coverings. We call this topology the weak G -topology.

Proposition 3.38. The weak G -topology induces the canonical topology of X .

Proof. Since the set of admissible open sets is the set of all affinoid subdomains, the result follows. □

By Proposition (3.26), any affinoid map $X \rightarrow Y$ between affinoid varieties is continuous with respect to the weak G -topology on X and Y . So if we take $\mathcal{C}$ to be sets of the form $\mathrm{Sp} A$ with their weak G -topology and morphisms as affinoid maps, Proposition (3.37) implies that there exists a unique finest G -topology $\mathfrak{T}'$ that is slightly finer than the topology on $\mathcal{C}$. For an affinoid variety X , we call the G -topology $\mathfrak{T}'_X$ the strong G -topology on X . Clearly, the weak G -topology satisfies (G_0) (take the identity map for $X = \mathrm{Sp} A$ and the zero ring for $\emptyset$). So the strong G -topology satisfies (G_0) , (G_1) and (G_2) . Also note that affinoid maps are continuous with respect to the strong G -topology by definition of $\mathcal{C}$ since morphisms are taken to be continuous maps.

By Proposition (3.38), we see that $\mathcal{O}_X$ becomes a sheaf with respect to the weak G -topology on X when we consider the category of admissible open subsets of X with morphisms as inclusions (for a categorical definition of a sheaf, see [Lei14]). In a similar manner, one can introduce a sheaf on a G -topological space, which enables us to define locally ringed spaces with respect to G -topologies similar to the ordinary topology. We give the definition below.

Definition 3.39. Let X be a G -topological space and let $\mathcal{O}_X$ be a sheaf of rings. The pair $(X, \mathcal{O}_X)$ is called a locally G -ringed space if the stalk $\mathcal{O}_{X,x}$ is a local ring for all $x \in X$.

We can also define morphisms between locally G -ringed spaces much like in the ordinary case. Namely, a morphism $(X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a pair (ϕ, ϕ^*) such that $\phi : X \rightarrow Y$ is continuous (in the sense of G -topology) and $\phi_V^* : \mathcal{O}_Y(V) \rightarrow \mathcal{O}_X(\phi^{-1}(V))$ is a (ring) homomorphism, where V runs over admissible open sets in Y , such that the diagram

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi_V^*} & \mathcal{O}_X(\phi^{-1}(V)) \\ \rho_{V,V'}^Y \downarrow & & \downarrow \rho_{\phi^{-1}(V), \phi^{-1}(V')}^X \\ \mathcal{O}_Y(V') & \xrightarrow{\phi_{V'}^*} & \mathcal{O}_X(\phi^{-1}(V')) \end{array}$$

commutes for each pair $V' \subset V$, where ρ^X and ρ^Y are the restriction maps, such that the stalk map $\phi_x^* : \mathcal{O}_{Y,\phi(x)} \rightarrow \mathcal{O}_{X,x}$ map the maximal ideal of $\mathcal{O}_{Y,\phi(x)}$ into the maximal ideal of $\mathcal{O}_{X,x}$ for all $x \in X$.

Let X be an affinoid variety. The sheaf $\mathcal{O}_X$ extends uniquely to a sheaf of k -algebras, also denoted by $\mathcal{O}_X$, with respect to the strong G -topology on X (see 9.2.3/1, [BGR84]). Then $(X, \mathcal{O}_X)$ is a locally G -ringed space (see 7.3.2/1, [BGR84]). Using this result, we view a k -affinoid variety as a locally G -ringed space over k . Moreover, a locally G -ringed space over k isomorphic to such a space is called a k -affinoid variety. Then k -affinoid maps now become k -morphisms of locally G -ringed spaces, written as (ϕ, σ) . We also change our notation from X to $(X, \mathcal{O}_X)$.

After this long discussion, we are now in a position to define rigid analytic varieties:

Definition 3.40. A rigid analytic variety over k , or a k -analytic variety, is a locally G -ringed space $(X, \mathcal{O}_X)$ over k such that the G -topology on X satisfies (G_0) , (G_2) and (G_3) and X admits an admissible covering $\{X_i\}_{i \in I}$ such that $(X_i, \mathcal{O}_{X_i})$ is a k -affinoid variety for each $i \in I$. A morphism of rigid analytic varieties $(X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ over k is a k -morphism of locally G -ringed spaces $(\phi, \sigma) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$.

4 Conclusion

The construction of rigid analytic varieties provides a resolution to the topological obstacles inherent in non-Archimedean fields. By shifting the geometric foundation from classical open sets to algebraically defined affinoid subdomains, we bypass the issue of total disconnectedness. The Tate algebra acts as the fundamental bridge, translating convergent power series into the algebraic language of maximal ideals and finite field extensions. Furthermore, the implementation of Grothendieck topologies ensures that local analytic data, governed by Tate's Acyclicity Theorem, can be coherently glued together globally. As discussed, the transition from the weak G -topology to the saturated strong G -topology preserves the continuity of morphisms while expanding the class of admissible sets. Ultimately, the category of k -affinoid varieties embeds fully and faithfully into the category of locally G -ringed spaces [BGR84], establishing rigid analytic varieties as the environment for global non-Archimedean geometry.

With this robust analytic category in place, a natural bridge to algebraic geometry emerges. Just as Serre's celebrated GAGA theorems establish an equivalence between complex projective varieties and complex analytic spaces, a remarkable analog holds in the non-Archimedean setting. The analytification of a proper algebraic scheme over k yields a rigid analytic variety whose category of coherent sheaves is entirely equivalent to its algebraic counterpart. Thus, the rigid analytic framework constructed here does not merely stand as an isolated analytic theory; it seamlessly intertwines with classical algebraic geometry, providing an essential machinery for modern advancements in arithmetic geometry.

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